

TASKS'

1.Open Command Prompt or Terminal and use the ping command to test connectivity to www.spotify.com. Note the number of packets sent, received, and the average response time.



```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>ping www.spotify.com

Pinging atc.spotify.map.fastly.net [151.101.211.42] with 32 bytes of data:
Reply from 151.101.211.42: bytes=32 time=81ms TTL=55
Reply from 151.101.211.42: bytes=32 time=74ms TTL=55
Reply from 151.101.211.42: bytes=32 time=103ms TTL=55
Reply from 151.101.211.42: bytes=32 time=32ms TTL=55

Ping statistics for 151.101.211.42:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 32ms, Maximum = 103ms, Average = 72ms

C:\Users\HP>
```

The connectivity to www.spotify.com was successfully tested using the ping command. 4 packets were sent and 4 packets were received, with 0% packet loss. The average response time was 72 ms.

2. Use the traceroute (tracert on Windows, traceroute on Linux/Mac) command to trace the path from your computer to www.flipkart.com. List the number of hops and identify the IP address of the first and last hop



```
C:\WINDOWS\system32\cmd. x + v
C:\Users\HP>tracert www.flipkart.com

Tracing route to e348826.dsca.akamaiedge.net [184.86.248.177]
over a maximum of 30 hops:

  1    3 ms    1 ms    3 ms  192.168.0.1
  2   23 ms    1 ms    5 ms  reliance.reliance [192.168.29.1]
  3   10 ms    5 ms   10 ms  10.9.32.1
  4   14 ms   32 ms   14 ms  172.31.3.177
  5   10 ms   16 ms    7 ms  192.168.103.178
  6    6 ms    9 ms    6 ms  172.26.100.229
  7    9 ms   17 ms   13 ms  172.26.100.210
  8   30 ms    7 ms   10 ms  192.168.188.54
  9    *      *      *    Request timed out.
 10    *      *      *    Request timed out.
 11    *      *      *    Request timed out.
 12   18 ms   14 ms   15 ms  a184-86-248-1.deploy.static.akamaitechnologies.com [184.86.248.1]
 13   20 ms   22 ms   16 ms  a184-86-248-177.deploy.static.akamaitechnologies.com [184.86.248.177]

Trace complete.

C:\Users\HP>
```

The traceroute command successfully traced the path from the computer to www.flipkart.com. The trace completed in 13 hops. The IP address of the first hop is 192.168.0.1, and the last hop is 184.86.248.177. Some intermediate hops (9–11) did not respond and showed “Request timed out.”

3. Simulate a connectivity issue by pinging a non-existent website (for example, ping `www.fakezomato12345.com`) and write down the error message you receive.
Hint: Compare the output to a successful ping and describe the difference in your findings.



```
C:\WINDOWS\system32\cmd. x + v
C:\Users\HP>ping www.fakezomato12345.com
Ping request could not find host www.fakezomato12345.com. Please check the name and try again.
C:\Users\HP>
```

The screenshot shows a Windows Command Prompt window with the title bar 'C:\WINDOWS\system32\cmd. x + v'. The command prompt displays the command 'C:\Users\HP>ping www.fakezomato12345.com' and the resulting error message: 'Ping request could not find host www.fakezomato12345.com. Please check the name and try again.' The prompt then shows 'C:\Users\HP>' on a new line. The taskbar at the bottom shows the Windows logo, a search bar, and several application icons including Edge, File Explorer, Spotify, and WhatsApp. The system tray on the right shows the date and time as '09:16 AM 25-09-2026'.

The ping was unsuccessful because the specified domain name could not be resolved. Unlike the successful ping to `www.spotify.com`, no reply or packet statistics were received. This indicates a DNS/name resolution problem for the non-existent domain.

**4.Imagine you are unable to access www.youtube.com from your system.
Use both ping and traceroute commands to diagnose the issue and write a
short report describing where the connection fails and what the possible
reasons could be.**



```
C:\WINDOWS\system32\cmd. x + v
C:\Users\HP>ping www.youtube.com

Pinging www.youtube.com [142.251.151.4] with 32 bytes of data:
Reply from 142.251.151.4: bytes=32 time=21ms TTL=111
Reply from 142.251.151.4: bytes=32 time=22ms TTL=111
Reply from 142.251.151.4: bytes=32 time=27ms TTL=111
Reply from 142.251.151.4: bytes=32 time=20ms TTL=111

Ping statistics for 142.251.151.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 20ms, Maximum = 27ms, Average = 22ms

C:\Users\HP>tracert www.youtube.com

Tracing route to www.youtube.com [142.251.151.4]
over a maximum of 30 hops:

  0  6 ms   4 ms   3 ms  192.168.0.1
  1  94 ms   2 ms  36 ms  reliance.reliance [192.168.29.1]
  2  67 ms  47 ms 352 ms 10.9.32.1
  3  25 ms   6 ms   9 ms 172.31.3.177
  4  17 ms   9 ms   7 ms 192.168.103.178
  5   7 ms   5 ms  24 ms 172.26.100.229
  6  15 ms   8 ms   5 ms 172.26.100.210
  7  18 ms  11 ms   7 ms 192.168.188.50
  8  *      *      *      Request timed out.
  9  *      *      *      Request timed out.
 10  *      *      *      Request timed out.
 11 26 ms   *      29 ms 142.251.151.4

Trace complete.
```

The ping test to www.youtube.com was successful, with 4 packets sent and 4 packets received and 0% packet loss. The average response time was 22 ms. The traceroute reached the destination in 11 hops. Hops 9 and 10 showed “Request timed out,” but the trace continued and successfully reached the destination at 142.251.151.4. Therefore, the timeout at hops 9 and 10 does not indicate that the connection failed; those intermediate devices may simply not respond to traceroute probes.